Portfolio Analyzer & Research Generator

AICA LEVEL 2 CAPSTONE PROJECT DOCUMENTATION

1. EXECUTIVE SUMMARY & CORE IDEA

The **Portfolio Analyzer & Executive Research Report Generator** was built to automate the end-to-end transformation of unformatted, multi-broker transaction exports (CSVs, Excel files) into institutional-grade, publication-ready PDF valuation reports. It eliminates manual ticker searching, date adjustments, and formatting overhead.

The Challenge

Financial advisors and portfolio managers spend excessive hours extracting broker exports, finding historical valuations, formatting tables, and gathering qualitative company research manually.

The Core Solution

An automated utility that parses messy inputs, resolves ticker/ISIN variants, fetches historical market trends, runs qualitative AI research, and renders typeset PDFs with custom branding.

2. KEY TOOLS & TECHNOLOGICAL STACK

Category	Tool / Library	Project Implementation Purpose
GUI Engine	Tkinter / ttk	Cross-platform desktop application interface with real-time logs.
Market Data API	yfinance	Fetches historical stock prices, closing metrics, and 12-week trends.
Symbol Matcher	Requests / Urllib	Resolves non-standard exchange symbols and ISINs to ticker symbols.
Data Processing	Pandas / NumPy	Parses raw CSV/XLSX files, cleans missing fields, and computes weights.
AI Integration	Gemini / OpenAI API	Generates qualitative insights (Governance, Financials, Sentiment).
PDF Generation	ReportLab	Constructs styled PDF reports with dynamic canvas pagination.

3. UTILITY ADVANTAGES & KEY INNOVATIONS

1. Robust SME Symbol Resolver

Handles non-standard suffixes (e.g., `-SM`, `-ST`) and small-cap Indian equities that normally break standard financial API search requests.

2. Historical Valuation Engine

Allows point-in-time valuation (e.g., "31 March 2026"). Calculates asset values, closes, and 12-week historical trend lines relative to any target date.

3. Graceful AI Fallbacks

Uses LLMs (Gemini/GPT) for stock summaries when API keys exist; seamlessly falls back to structured rule-based templates if offline or unkeyed.

4. Flexible Column Parser

Auto-identifies variant headers like `Scrip`, `Particulars`, `Qty`, `Buy Price`, and `CMP` without requiring pre-formatting.

4. DEVELOPMENT ITERATIONS & EVOLUTION

Phase 1: Ingestion & Historical Engine
Built flexible file parsers (.csv, .xlsx) and integrated yfinance date-slicing logic.

Phase 2: Typesetting & PDF Generation
Implemented ReportLab canvas pagination, metric callouts, and 12-week price trend tables.

Phase 3: SME Fixes & Edge Case Handling
Added flexible date parsers and custom mapping overrides for restricted and SME tickers.

Phase 4: UI Refinement & Polish
Integrated dynamic client naming, contrast-adjusted headers, and non-blocking multi-threading.

5. PIPELINE ARCHITECTURE OVERVIEW

